How to write and debug SQL queries

SQL queries have a couple properties which make writing and debugging different than in conventional programming languages.

- SQL queries are declarative. You have less control over what operations are done and what order.
- SQL queries have the source table names hard-coded. They are not like functions or procedures where you can easily pass different data.

However, we can adapt standard strategies to SQL.

Writing SQL queries

- Use good style -- Write your queries with good code style that is easy to understand.
- Solve a simpler problem first -- Simplify the problem, solve it, then adapt the solution to the original problem.
- Decompose the problem -- Break down the problem into simpler parts. Develop complex queries as multiple, simpler queries. Combine the parts into one query via subqueries or WITH.
- Don't optimize until you have a working solution -- Optimization often results in less understandable code. Get a working version first. If that's inefficient, optimize.

Debugging SQL queries

- Test on multiple inputs -- This requires you to change the contents of the source tables. Be sure to test special cases, such as NULL data and empty tables.
- "Print" more data -- We can't simply insert print statements in the middle of a query. Instead, the rough equivalent is to add fields or computed values to the SELECT clause, since we often print query results.
- Create or use a unit testing framework -- This is somewhat more complicated and also less common than for conventional programming languages, but still uses the same basic steps:
    - Create sample data. Load it into the desired tables.
    - Run the query. Store the results in a table.
    - Compare the actual results to what was expected. Report discrepancies.